

TANUSH BAMNOTE

Nagpur, Maharashtra

+91 9284443508 | tanushbamnote2@gmail.com | linkedin.com/in/tanush-bamnote | github.com/7H4N05

PROFESSIONAL SUMMARY

Full-stack software engineer and cybersecurity specialist experienced in backend architecture, malware analysis, threat detection, and scalable application development. Skilled in Python, TypeScript, React, FastAPI, Docker, and offensive security techniques with expertise in building production-grade systems and security operations platforms.

EDUCATION

B.Tech in Computer Science Engineering (Cyber Security)
St. Vincent Pallotti College of Engineering and Technology

Expected 2027
CGPA: 8.60

TECHNICAL SKILLS

Backend Development: FastAPI, Python, Node.js, REST APIs, GraphQL, Asynchronous Programming, Microservices Architecture

Frontend Development: React 18, TypeScript, Tailwind CSS, Zustand, Tauri (Desktop), HTML5, CSS3

Databases: SQLite, MySQL, PostgreSQL, SQLAlchemy, Async Query Handling

Cloud and DevOps: Docker, Docker Compose, AWS EC2, Linux Administration, Git, GitHub, CI/CD

Cybersecurity: Malware Analysis, Reverse Engineering, Web Exploitation, Vulnerability Assessment, Digital Forensics, Incident Response, Threat Detection, Penetration Testing, Security Automation

Security Tools: Burp Suite, Metasploit, Nmap, Wireshark, Ghidra, JADX, Apktool, Androguard, Pyrogram

Programming Languages: Python, TypeScript, JavaScript, C++, PHP, Bash, Perl

Tools and Platforms: Prometheus, Grafana, Psutil, PIL, OpenAPI, Figma, XAMPP

EXPERIENCE

Cyber Police Nagpur | Cybersecurity Intern

July 2025 to August 2025

- Assisted in 300+ cybercrime investigations involving phishing, fraud, and malware incidents.
- Reverse engineered 25+ malicious Android APKs using Ghidra, JADX, and static analysis techniques.
- Contributed to development of APK Inspector, improving malware investigation workflows.
- Applied malware analysis, incident response, and penetration testing techniques to identify and mitigate threats.

Website Developer & Security Infrastructure Engineer

October 2025 to November 2025

Phoenix CyberSecurity Club, Amgaon

- Designed and deployed official club website using MERN Stack (React, Node.js, MongoDB) on AWS and Docker.
- Implemented secure authentication, dynamic event management, CTF integration, and responsive UI with Tailwind CSS.
- Containerized deployment achieving 99% uptime with automated CI/CD pipelines and Linux administration.

PROJECTS

1. TeleGallery | Production Grade Telegram Media Upload Platform

[Link]

- Built scalable production system handling 500GB+ uploads with resumable transfers, automated batching, and multi-account failover.
- Architected resilient upload orchestration using FastAPI, Pyrogram, SQLite, and Docker with crash recovery, exponential retry backoff, and adaptive FloodWait handling.
- Implemented real time monitoring dashboard using React 18, TypeScript, Zustand, and Tailwind CSS with live SSE event streaming.
- Designed self-healing reliability layer with health monitoring, failure detection, adaptive congestion control (AIMD), and ETA analysis achieving 99.7% uptime.
- Engineered Prometheus metrics exporter and Grafana integration for production monitoring and performance analysis.

2. SOC Platform | Security Operations Center

[Link]

- Designed microservices architecture for centralized security monitoring with real time threat detection and incident response workflows.
- Developed event ingestion pipeline, threat analysis dashboards, and automated alerting system using Python and modern web technologies.
- Implemented security event aggregation, log correlation, real time alert management, and automated response capabilities.

3. RiskShield | Risk Assessment and Vulnerability Management Platform

[Link]

- Built comprehensive security risk assessment platform for vulnerability management and mitigation tracking.
- Integrated automated risk scoring, security analytics, and compliance reporting using Python and data analysis tools.

4. Traffic Police Management System | Full Stack Web Application

[Link]

- Built comprehensive web application for traffic management with vehicle registration, violation tracking, and payment processing.

- Designed relational database schema with 4 core tables ensuring data persistence and integrity with user authentication and role based access control.
 - Tech stack: PHP 7.4+, MySQL 5.7+, HTML5, CSS3 with XAMPP deployment.
- 5. APK Inspector | Android Malware Analysis Platform** *(Internal Project)*
- Built full stack malware analysis platform automating APK decompilation, static detection, and forensic reporting using Apktool, JADX, Androguard with FastAPI backend.
- 6. CTF Infrastructure and Security Challenge Development** *(Event Infrastructure)*
- Designed and deployed scalable CTF infrastructure using Docker, AWS, and XAMPP supporting concurrent teams with backend challenge management and real time leaderboards.
- 7. PicoCTF | Security Challenges** [Link]
- Solved 50+ cybersecurity challenges covering cryptography, reverse engineering, web exploitation, and binary analysis.
- 8. Phoenix Website | Official Cybersecurity Club Platform** [Link]
- Designed and deployed official website for Phoenix Cybersecurity Club using MERN Stack with event management, member directory, and CTF integration on AWS with Docker.

ACHIEVEMENTS

- 1st Place, YCCE Capture The Flag (CTF) 2025
- 1st Place, GHRCOE Capture The Flag (CTF) 2025
- 2nd Place, HicaThon Capture The Flag (CTF), SSPU Pune
- 5th Place, DefSec Con Capture The Flag (CTF)
- Coordinator, EncipherX 2025. Led 24 hour national level CTF event with 500+ participants, 100+ teams, and 20% year over year growth. Designed infrastructure serving 250+ workshop participants. Managed team of 15+ organizers.
- Coordinator, Cyberpunk (Technical Event). Coordinated major college-level cybersecurity event featuring 4 sub-events for 200+ attendees.
- Coordinator, Hackers Heist (College-Level CTF). Hosted 5hr solo CTF for 100+ participants and conducted Wi-Fi hacking & Linux security workshops for 200+ students.

LEADERSHIP

- **Admin, Phoenix Cybersecurity Club:** Oversee 100+ active members, organize monthly CTF competitions, technical workshops, training sessions, and community engagement initiatives.
- **General Secretary, SADC:** Coordinate technical initiatives, development workshops, organize security awareness campaigns, and foster industry collaboration programs.